

PROLOG PROGRAM FOR FRUITS AND THEIR COLOURS USING BACKTRACKING

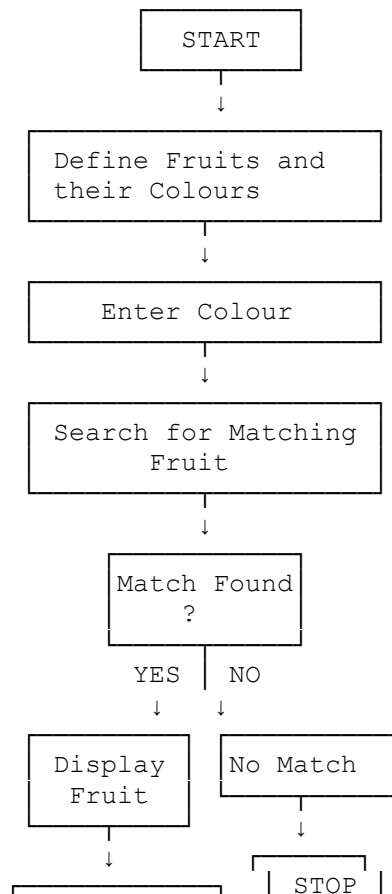
Aim

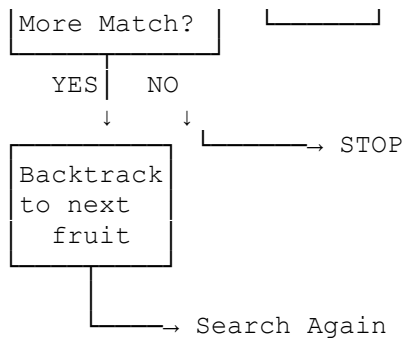
To write a Prolog program to identify fruits and their corresponding colours using backtracking.

Algorithm

1. Start.
2. Define facts for different fruits and their colours.
3. Create a predicate to find a fruit based on its colour.
4. Enter a colour as input.
5. Prolog searches for a fruit matching the given colour.
6. If multiple fruits match, use ; **(semicolon)** for backtracking.
7. Display all matching fruits.
8. Stop.

Flowchart





Prolog Program

% Fruit and Colour using Backtracking

```
fruit(apple, red).  
fruit(banana, yellow).  
fruit(orange, orange).  
fruit(grapes, green).  
fruit(mango, yellow).
```

```
find_fruit(Color, Fruit) :-  
    fruit(Fruit, Color).
```

Output

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)  
File Edit Settings Run Debug Help  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.  
  
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).  
  
?-  
% c:/Users/Madhu Bavana/OneDrive/Desktop/Pr1/Fruit and Colour using Backtracking.pl compiled 0.00 sec, 6 clauses  
?-  
| find_fruit(yellow, Fruit).  
Fruit = banana .  
?- find_fruit(red, Fruit).  
Fruit = apple.  
?-
```

Explanation

The query searches for fruits having the colour **yellow**. Prolog first finds **banana**. When ; is pressed, Prolog **backtracks** and finds the next matching fruit, **mango**.